

AP CALCULUS AB - UNIT 4

Guided Notes 4.3: Other Rates of Change



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we calculate and interpret derivatives in scientific, economic, and geometric contexts?

- Calculate and interpret derivatives in scientific, economic, and geometric contexts.
- Distinguish a modeled quantity from its instantaneous rate of change.
- Use correct units and signs in applied interpretations.
- Combine derivative rules to calculate rates for derived quantities.

Key Knowledge, Vocabulary, and Notation

Rate in context

For $Q = Q(t)$, $Q'(t)$ describes output change per unit input.

Marginal quantity

For cost, revenue, or profit as a function of production, a derivative approximates change caused by one additional unit.

Sensitivity

If $Q = Q(x)$, then $Q'(a)$ measures how sensitive Q is to a small change in x near a .

Relative rate

$\frac{Q'(t)}{Q(t)}$ gives fractional change per input unit; multiply by 100% for percent rate.

Derived rate

Differentiate the formula defining a derived quantity, then substitute the required input.

Units check

Use dimensional analysis to detect impossible interpretations or algebraic errors.

Concept Development and Strategy

1. Begin by naming the independent and dependent variables and their units.
2. A derivative can be computed symbolically, estimated from data, or read as slope from a graph; the interpretation format remains the same.
3. Do not confuse a large amount with a large rate. Compare rates using derivative values and magnitudes.
4. For marginal interpretations, say 'approximately' because a derivative predicts a small discrete change.
5. When a new quantity is built from several changing quantities, apply the appropriate product, quotient, or Chain Rule.

Shared Representation / Data

t (hours)	0	2	4	6	8
$W(t)$ (kW)	18.0	22.4	27.1	31.0	33.2
$W'(t)$ (kW/hour)	2.40	2.05	1.55	0.80	0.15

Worked Example: Marginal profit

Profit is $P(q) = 70q - 0.04q^2 - 5000$ dollars. Find and interpret $P'(600)$.

Answer and Reasoning

$P'(q) = 70 - 0.08q$, so $P'(600) = 22$. Near 600 units, one additional unit increases profit by approximately \$22.

Worked Example: Geometric sensitivity

The area of a circle is $A(r) = \pi r^2$. Find and interpret $A'(10)$.

Answer and Reasoning

$A'(r) = 2\pi r$, so $A'(10) = 20\pi$ cm²/cm. Near radius 10 cm, area increases by about 20π cm² per additional centimeter of radius.

Worked Example: Relative growth

A culture satisfies $N(t) = 800e^{0.12t}$. Find its relative growth rate.

Answer and Reasoning

$N'/N = 0.12$, so the instantaneous relative growth rate is 12% per time unit.

Worked Example: Rate from a table

Using the shared table, interpret $W'(6) = 0.80$.

Answer and Reasoning

At hour 6, electrical power usage is increasing at 0.80 kilowatt per hour.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. The radius r is in cm and area A in cm^2 . State units of dA/dr .

2. Interpret $C'(250) = 8.60$ when C is cost in dollars and q is items.

3. Volume is $V(r) = \frac{4}{3}\pi r^3$. Find $V'(5)$.

4. Demand is $D(p) = 1200 - 30p$. Interpret $D'(p)$.

5. A rectangular region has $A(t) = L(t)W(t)$. If $L = 8$, $W = 5$, $L' = 0.3$, and $W' = -0.2$, find A' .

6. A company's revenue is $R(q) = q(120 - 0.02q)$. Find output levels where marginal revenue is positive.

7. A quantity Q has units joules and depends on temperature T in kelvins. What are the units and meaning of d^2Q/dT^2 ?

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. If $A(r)$ is area in m^2 and r is meters, units of $A'(r)$ are

- | | |
|------------------|------------------|
| (A) m^3 | (B) m^2 |
| (C) m | (D) $1/\text{m}$ |

2. Marginal cost $C'(q)$ is best interpreted as

- | | |
|--|---------------------------|
| (A) total cost | (B) average cost per unit |
| (C) approximate change in cost for one more unit | (D) profit per unit |

3. For $P(q) = 80q - 0.03q^2 - 4000$, marginal profit at $q = 900$ is

- | | |
|--------|--------|
| (A) 16 | (B) 26 |
| (C) 53 | (D) 80 |

Common Errors and AP Precision

- State the quantity, sign, input value, and units in every contextual interpretation.
- In related rates, differentiate before substituting values. In linearization, identify the base point. For L'Hopital's Rule, verify $0/0$ or ∞/∞ first.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

Suppose $R(p) = pD(p)$ and demand decreases with price. Derive a condition for revenue to increase with price.

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